

12.112

AI25BTECH11012 - GARIGE UNNATHI

Question:

If the entries in each column of a square matrix $\mathbf{M}$ add up to 1, then an eigenvalue is

- 1) 4
- 2) 3
- 3) 2
- 4) 1

Solution:

The question states that the sum of all elements in each column of the square matrix $\mathbf{M}$ is 1 then we have to find a number which is an eigen value of such matrix

If we take a matrix

$$\mathbf{v} = (1, 1, \dots, 1)^T \quad (4.1)$$

Then , we can say that

$$\mathbf{M}\mathbf{v} = \mathbf{v} \quad (4.2)$$

From the above equation we can say that one of the eigen value for $\mathbf{M}$ is 1
Hence option 4 is true